Scope-VSR: A Broad-Coverage Training Dataset for Video Super-Resolution

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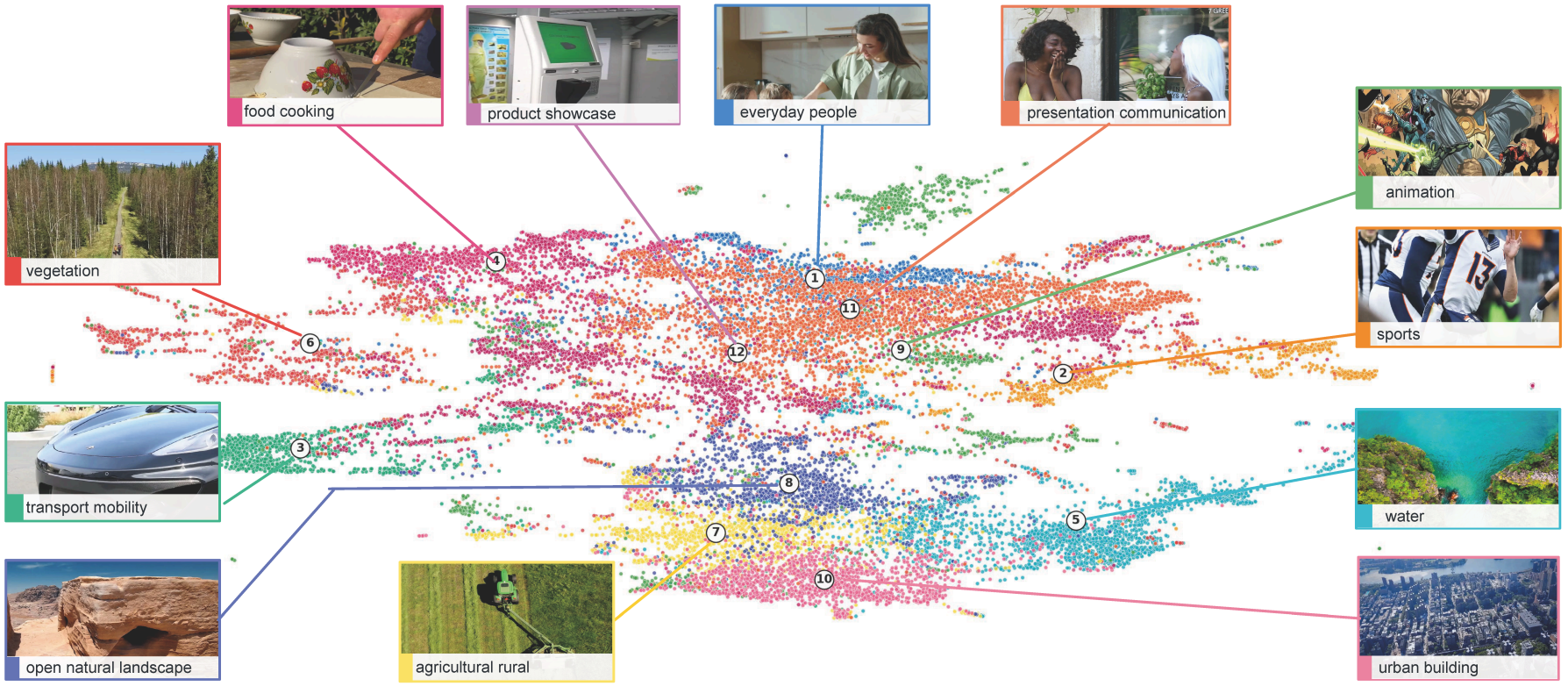
Under Review

Paper | arXiv | Supplementary | Code

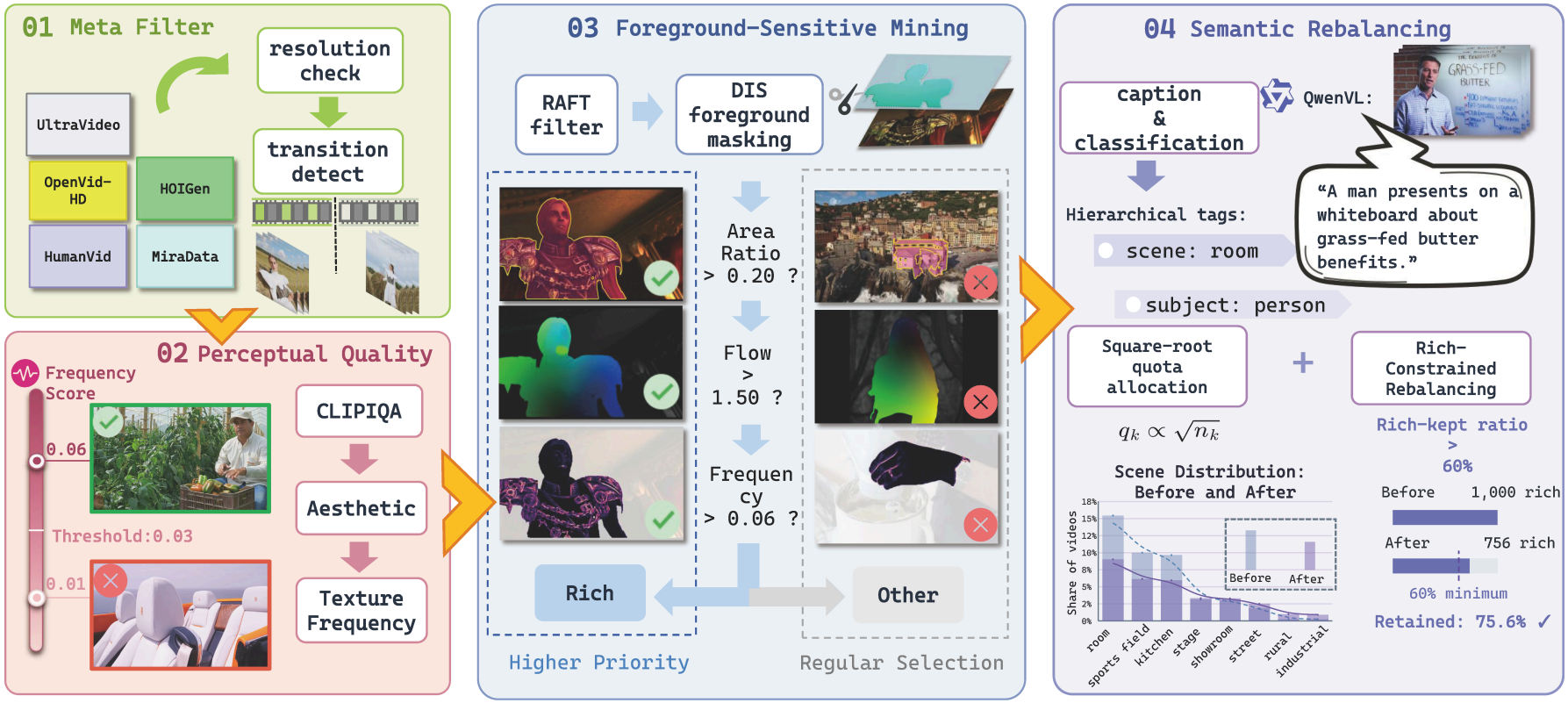


The semantic atlas of SCOPE-VSR. Colors denote topics, with representative frames illustrating the diverse semantic content of our dataset.

Abstract

Diffusion models have shown superior performance in video super-resolution (VSR). However, restoring diverse real-world content remains challenging due to the lack of balanced semantic coverage in existing training data. In this paper, we introduce a structured curation pipeline for open-domain VSR, SCOPE. In this pipeline, we combine task-oriented video selection with hierarchical semantic rebalancing. Applying this pipeline to a large-scale collection of data sources, we construct the SCOPE-VSR dataset for training and SCOPE-test for testing. SCOPE-VSR has approximately 27k videos with a balanced and diverse coverage of various real-world scenarios and subjects. In addition, we propose GraCo-VSR, a novel framework for one-step video restoration. GraCo-VSR adopts a reliability-aware design with explicit supervision to reduce degradation-induced distortions in motion and texture guidance. This design enables the model to better leverage the diverse training content in SCOPE-VSR, thereby supporting generalization across varied scenarios. Experiments demonstrate that our model exhibits strong performance on both perceptual fidelity and visual quality.

Method



Overview of the SCOPE data construction pipeline

We propose SCOPE, a systematic pipeline that combines multi-stage curation with semantic rebalancing. We collect approximately 0.6 million video clips from multiple domains, including OpenVid-HD, UltraVideo, MiraData, HumanVid, and HOIGen. The original collection contains. By applying the pipeline, we extract 27,351 clips to form the final SCOPE-VSR dataset.

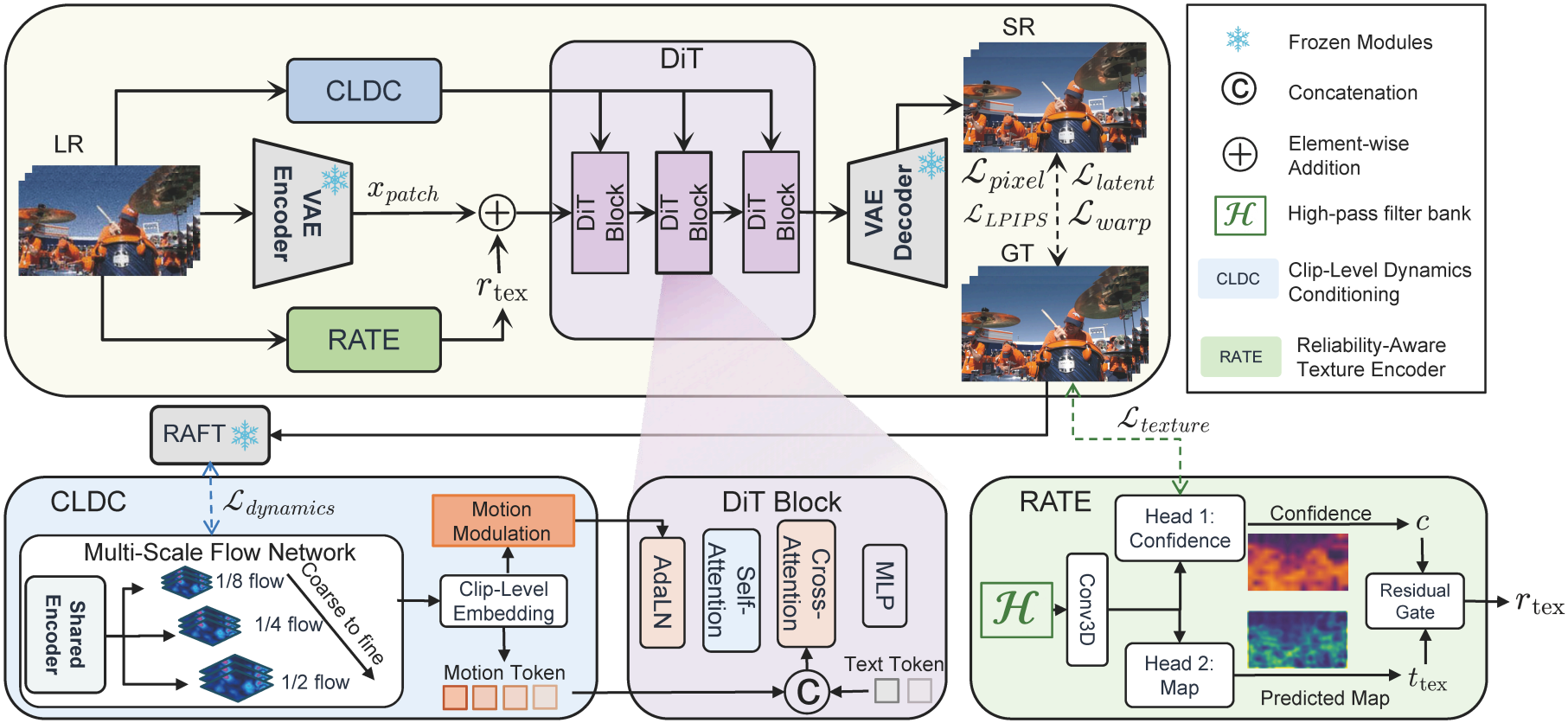


Illustration of the GraCo-VSR framework

We introduce GraCo-VSR, a one-step video restoration framework which provides reliable and granularity-aligned guidance for open-domain VSR. The Reliability-Aware Texture Encoder (RATE) is designed to extract confidence-weighted local texture cues, while the Clip-Level Dynamics Conditioning (CLDC) is designed to provide robust clip-level motion guidance.

Results

▼ Quantitative Results (click to expand)

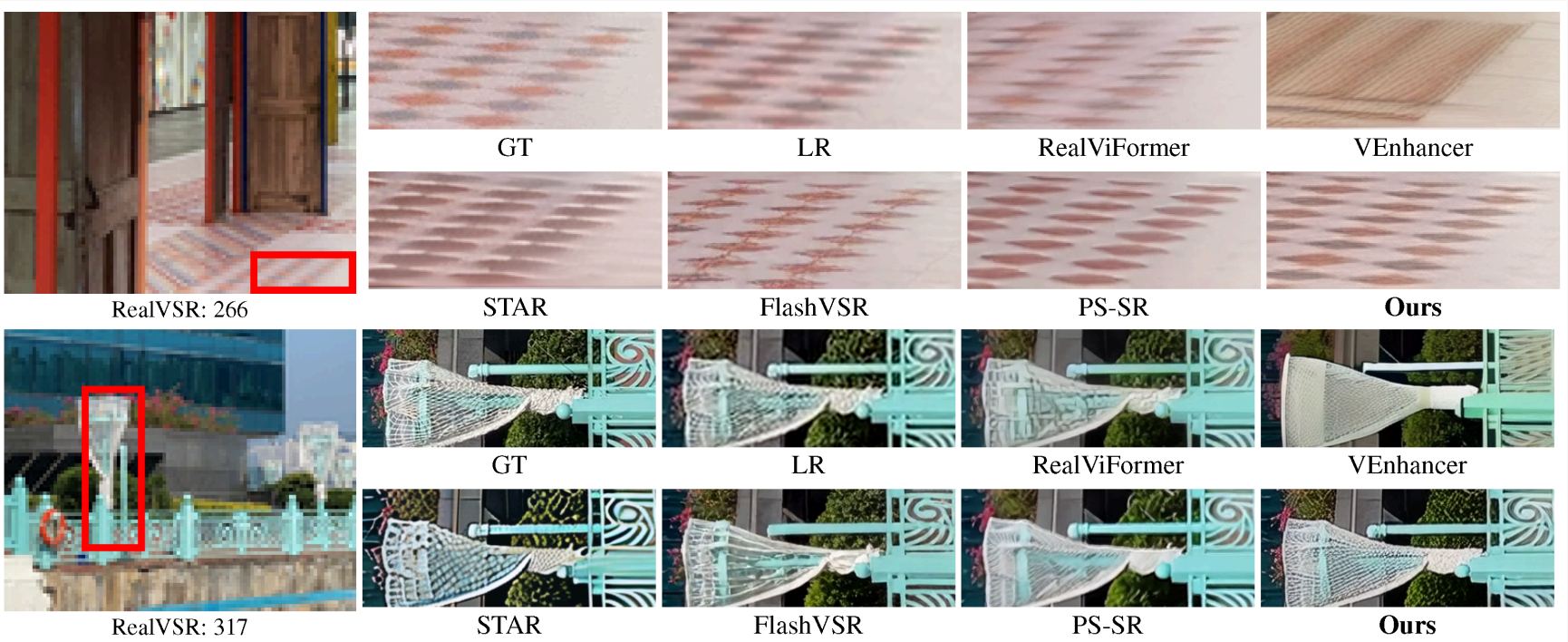
- Quantitative results on synthetic and real-world datasets (Tab. 1 of the main paper)

Dataset	Metric	RealBasicVSR	RealViFormer	VEnhancer	STAR	FlashVSR	PS-SR	**GraCo-VSR (ours)**
UDM10	PSNR ↑	24.0000	24.6400	21.3134	23.9995	23.4229	23.3345	**25.2523**
	SSIM ↑	0.6968	0.6947	0.6803	0.6983	0.6951	0.7186	**0.7263**
	LPIPS ↓	0.3636	0.3681	0.4335	0.4129	0.2956	0.3075	**0.2558**
	DISTS ↓	0.2012	0.2039	0.2305	0.2213	**0.1564**	0.1600	0.1673
	CLIP-IQA ↑	0.4377	0.4157	0.2852	0.2410	0.5146	0.4120	**0.5197**
	MUSIQ ↑	58.3230	57.9000	37.2538	36.6531	**65.2129**	62.1139	62.0742
	MANIQA ↑	0.2829	0.2878	0.2120	0.2122	**0.3767**	0.3320	0.3153
	DOVER ↑	0.7343	0.7303	0.4686	0.4780	0.7657	0.7566	**0.7707**
YouHQ40	PSNR ↑	22.0202	**23.6600**	19.7806	22.5653	21.8712	22.1063	23.2283
	SSIM ↑	0.5839	**0.6334**	0.5911	0.6281	0.5818	0.6045	0.6266
	LPIPS ↓	0.4048	0.3662	0.4742	0.4655	0.2991	0.3570	**0.2670**
	DISTS ↓	0.1891	0.1940	0.2140	0.2361	**0.1354**	0.1614	0.1680
	CLIP-IQA ↑	0.4729	0.4462	0.3309	0.2762	**0.5071**	0.4283	0.4922
	MUSIQ ↑	62.3136	63.1800	42.0159	33.1669	**68.8096**	63.6817	63.5833
	MANIQA ↑	0.2973	0.2903	0.2442	0.2346	**0.3833**	0.3434	0.3120
	DOVER ↑	0.8485	0.8464	0.6933	0.5968	0.8728	0.8510	**0.8763**
RealVSR	PSNR ↑	22.0039	**22.1600**	15.5017	17.7302	19.7673	20.8142	19.8605
	SSIM ↑	**0.7163**	0.6955	0.3934	0.5420	0.5632	0.6649	0.6258
	LPIPS ↓	0.2044	0.2023	0.3838	0.3151	0.2125	0.2261	**0.1973**
	DISTS ↓	**0.1018**	0.1138	0.1734	0.1667	0.1083	0.1182	0.1053
	CLIP-IQA ↑	0.3538	0.3108	0.4249	0.4724	0.4083	0.2986	**0.6054**
	MUSIQ ↑	70.2996	66.6800	72.3089	70.3364	71.9875	66.5488	**74.8212**
	MANIQA ↑	0.3802	0.3341	0.4078	0.3888	**0.4236**	0.3542	0.4161
	DOVER ↑	0.7369	0.7191	0.7731	0.7033	0.7590	0.7566	**0.8203**
REDS30	PSNR ↑	21.3318	21.7705	13.3273	21.5222	20.3816	21.2262	**21.6932**
	SSIM ↑	0.5201	0.5397	0.4094	0.5477	0.5033	**0.5605**	0.5432
	LPIPS ↓	0.3964	0.3732	0.7798	0.5631	0.3478	0.4060	**0.3345**
	DISTS ↓	0.1950	0.1893	0.4003	0.2635	**0.1600**	0.1880	0.1969
	CLIP-IQA ↑	0.3233	**0.3836**	0.2427	0.1866	0.3502	0.2858	0.3552
	MUSIQ ↑	**61.0596**	60.8387	23.8483	31.3451	60.9516	55.5433	55.5800
	MANIQA ↑	0.2868	0.2823	0.1636	0.1639	**0.2910**	0.2360	0.2388
	DOVER ↑	0.5812	0.5896	0.2485	0.3595	**0.6874**	0.5864	0.6243
SCOPE-test	PSNR ↑	25.1819	25.6788	20.8188	23.5462	24.1059	23.9382	**25.7405**
	SSIM ↑	0.7202	0.7278	0.6174	0.6827	0.7050	0.7267	**0.7427**
	LPIPS ↓	0.2916	0.2756	0.3903	0.3948	0.2215	0.2645	**0.2056**
	DISTS ↓	0.1707	0.1675	0.1830	0.1969	**0.1193**	0.1445	0.1406
	CLIP-IQA ↑	0.4274	0.3829	0.4113	0.3009	0.4715	0.3724	**0.4767**
	MUSIQ ↑	64.3304	62.9521	61.9197	44.8322	**68.5916**	64.0642	65.8728

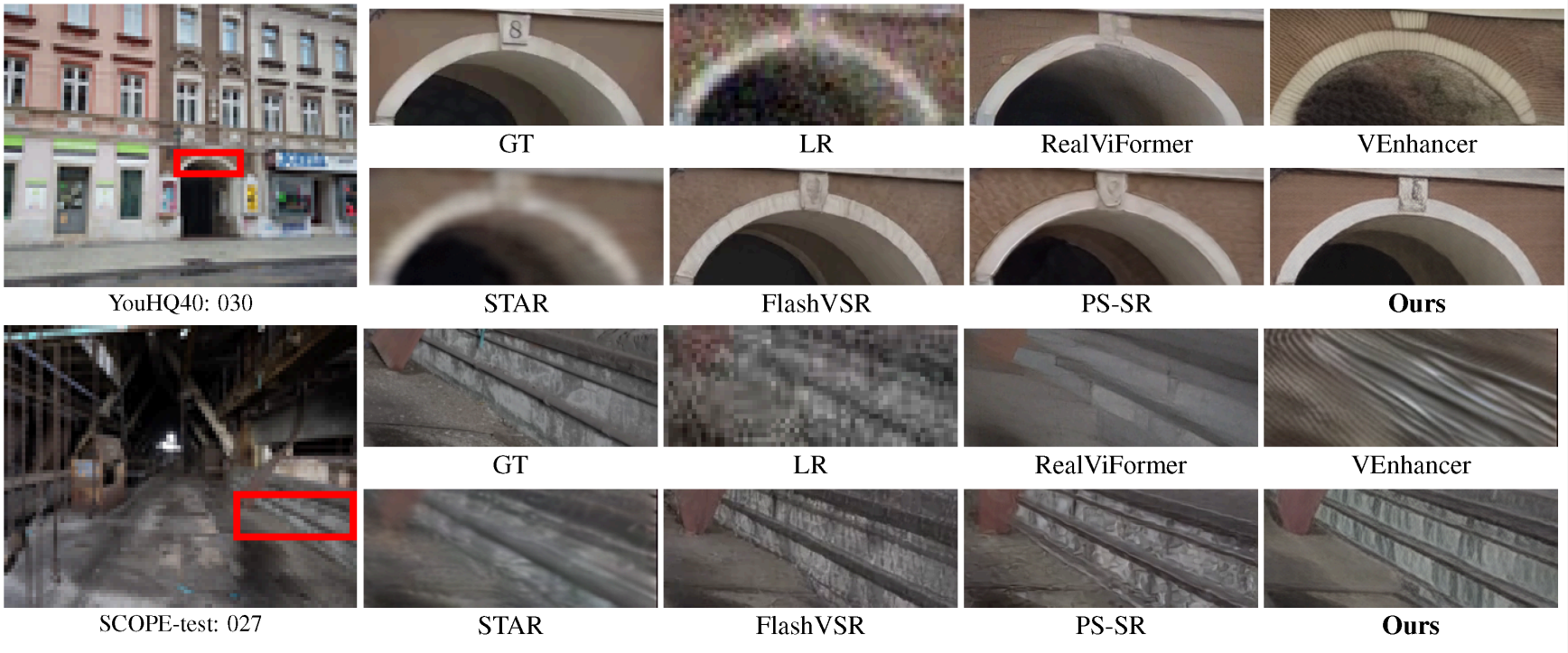
MUSIQ ↑	64.3304	62.9321	61.9197	44.8322	**68.3910**	64.0042	65.8728
MANIQA ↑	0.3140	0.2931	0.3030	0.2288	**0.3916**	0.3165	0.3253
DOVER ↑	0.7520	0.7187	0.7658	0.5828	**0.8024**	0.7522	0.7717

▼ Qualitative Results (click to expand)

- Visual comparisons on the real-world RealVSR dataset (Fig. 6 of the main paper)



- Visual comparisons on the synthetic YouHQ40 and SCOPE-test datasets (Fig. 7 of the main paper)



▶ More Results (click to expand)

BibTeX

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